BRAD BURCH

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SUMMARY

Senior full-stack software engineer with 7 years building production systems at Salesforce, Total Brain, Front, and Pawservation. Architected and own a booking, payments, and Claude agent platform (MCP, OAuth 2.1 + PKCE, Stripe Connect) on Cloudflare Workers, from data model and money invariants through evals, on-call, and incident response.

SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, Java, Scala, SQL, React 19, Node.js, Hono, Spring, Play
Infrastructure: AWS (DynamoDB, SQS), serverless (Cloudflare Workers, Durable Objects, Queues, Workflows), PostgreSQL, SQLite (D1), S3-compatible object storage (R2), KV, MSSQL, Docker, Kubernetes, REST APIs, webhooks, distributed systems
AI / LLM: Claude (Haiku, Sonnet), Vercel AI SDK, Model Context Protocol (MCP), tool calling, LLM evals, prompt caching, prompt-injection defense, AI cost metering, Claude Code, Cursor
Payments & Security: Stripe (Connect, Billing, Checkout), OAuth 2.1 + PKCE, JWT, SAML 2.0, HMAC webhook verification, AES-GCM, SSRF prevention, multi-tenant isolation
Testing & Delivery: Vitest (Node and Workers runtime), Playwright, contract tests, GitHub Actions, Jenkins, CI/CD, alerting

EXPERIENCE

Software Engineer 2022 - Present
Pawservation, San Francisco, CA (pawservation.com) | booking, payments, and AI platform for pet-care businesses

- Architected a production TypeScript booking and payments platform (React, Hono, Cloudflare Workers, D1, Durable Objects, Workflows) serving 30+ active clients, 60+ bookings/month, and 4+ outside pet-sitting businesses.
- Designed one set of shared tool definitions for a Claude Haiku 5.5 agent on web chat and WhatsApp plus a remote MCP server; with the client portal, more than half of bookings now arrive self-serve.
- Guaranteed each stay is auto-charged at most once by claiming a D1 uniqueness constraint before calling Stripe, verified with 20 concurrent writers on the real D1 runtime.
- Replaced date-ordered payment matching with closest-first allocation, ending a defect class that cleared a $600 stay with a $40 payment and flagged a $2,000 prepayment as delinquent.
- Hand-built an OAuth 2.1 + PKCE authorization server for the MCP server (S256, rotating hashed refresh tokens, SSRF-guarded client metadata), letting clients book directly from Claude.
- Cut agent cost 4.9x per eval run by restructuring the system prompt into a byte-stable cached prefix, serving about 95% of eval-suite input tokens from cache.
- Added change detection to a scheduled calendar sync, eliminating the source of 97% of daily database writes (56,424 of 58,121 against a 100,000/day platform cap).
- Owned reliability for unattended jobs (real-Workers-runtime CI, auth-failure alerts, completion markers), recovering four months of invoice runs by cutting per-PDF CPU time from 154 ms to 18 ms.
- Set the acceptance bar for AI changes with a 47-case live-model eval suite and a benchmark of 8 open-weight models, where Claude Haiku scored 100% at 2.1 s mean latency.

Software Engineer 2022
Front, San Francisco, CA (role eliminated in company-wide layoff)

- Shipped a Node.js REST API enabling account-level email rules for multi-domain enterprise customers, previously configurable only per-inbox.
- Shipped a security upgrade to the core JavaScript email library handling 200K+ outbound messages per month.

Software Engineer 2019 - 2021
Total Brain, San Francisco, CA

- Owned enterprise integration delivery as engineering point of contact on Customer Success and Sales calls, standardizing SAML 2.0 and JWT SSO plus HubSpot and Box integrations to cut onboarding from 2+ weeks to ~3 days.
- Migrated user data from MSSQL to DynamoDB with a redesigned NoSQL schema, cutting average API response times by 50%, and mentored junior engineers on backend patterns.
- Shipped a Scala backend integration for Apple, Google, and Facebook authentication, contributing to a 60% lift in new sign-ups.

Software Engineer 2017 - 2018
Salesforce, San Francisco, CA

- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams.
- Built an OAuth2 security filter for microservice access and a React management UI that cut internal test setup time 40%.

PROJECTS

- **MCP Auth Kit (open source):** OAuth 2.1 + PKCE kit for MCP servers with rate limits, scoped tools, and two-phase confirmation.
- **Debrief (open source):** macOS interview-coaching app with on-device WhisperKit transcription and Claude scoring (Swift).
- **Roadrunner (open source):** Django app matching eBird and iNaturalist sightings to Strava activities (Python, PostgreSQL).

EDUCATION

DePauw University, Bachelor of Arts in Computer Science 2017